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Model Risk Management and Validation Frameworks for Machine Learning Models in Banking: A Comprehensive Review

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Abstract
The infusion of machine learning (ML) into banking has revolutionized credit decisioning, fraud detection, and risk management, but the black-box nature of ML models poses unique challenges for traditional model risk management (MRM). This study provides an overview of recent developments (2020-2025) in MRM frameworks for ML models in banking, including regulatory changes, validation methods, explainability approaches, and upcoming governance issues. We critically examine peer-reviewed papers, regulatory guidelines, and industry reports to determine trends, challenges, and knowledge deficits in the validation and risk management of ML models. There are substantial discrepancies among traditional MRM frameworks, and ML models need prompting model interpretability, bias identification, continuous monitoring, and third-party model governance. Regulatory agencies are currently updating their guidelines, but uniform validation protocols remain in a gap. Explainable AI methods are promising but face scalability limitations. Novel ML model risk management will necessitate a fundamental rethink of validation frameworks that also include dynamic monitoring, fairness testing, strong governance structures, and interdisciplinary knowledge.

Keywords: Model Risk Management, Machine Learning, Model Validation, Explainable AI, Regulatory Compliance, Algorithmic Fairness

1. Introduction

Machine learning adoption in the banking sector has recently increased significantly, driven by superior prediction performance of ML techniques over traditional statistical models in credit risk assessment, fraud detection and operational risk management (Bhatore et al., 2020; Leo et al., 2019). Nonetheless, ML models are quite different from traditional statistical approaches in terms of black box properties, the ability to learn dynamically and sophisticated nonlinear pattern recognition (Fuster et al., 2022). The features uncover drawbacks of prevailing MRM systems for interpretable statistical learning.

The recent reports show that large financial institutions have actively implemented or are considering how to implement ML models with the use cases showing more accurate and operational efficiency (Shi et al., 2022). These risks were further highlighted during the Covid-19 pandemic, where pre-pandemic models of credit risk on which traditional ratings are based fell short by not taking into account what economists

called 'Black Swans' (Li et al., 2022), leading to a call for more robust assessments through new MRM frameworks in the global markets (Heß & Damásio, 2025).

However, despite the increasing recognition of these challenges, a holistic review of MRM frameworks across model development, validation, ongoing monitoring and governance for AI/ML systems is quite fragmented (Shi et al., 2022; Tian et al., 2024). The literature does not contain a structured assessment of how these traditional validation pillars need to be adjusted for the peculiar characteristics presented by ML. Further, state-of-the-art XAI approaches often do not focus on direct integration with practical regulatory needs and scalability in production settings (Misheva et al., 2021; de Lange et al., 2022).

This article offers a needed systematic synthesis of ML model risk management progress in banking (2020-2025), with a focus on:

1. Aggregating literature for MRM adaptations in ML/AI models.
2. Examining the state-of-the-art validation methods in relation to new ones based on XAI explanation techniques and bias noise detection formulations.
3. Examining regulatory evolution across jurisdictions.
4. Evaluating practical implementation challenges.
5. Identify limitations and recommend future areas of study.

We adopt expert-driven synthesis interpretations of complex developments in ML model risk management using a trend-based approach.

2. Emerging Trends and Thematic Analysis

2.1 Regulatory Framework Evolution

There has been considerable change in the regulatory environment. First, the UK's SS 1/23 (May 2023) which stands out as a landmark in the sense that it is the first regulation for banking specifically worded on AI/ML model risk, building on traditional MRM foundations and adding straight lines of sight into ML specifics such as: drift monitoring, carried over documentation and proportionate explainability expectation according to level of model risk being triggered (Bank of England, 2023).

The European Union's (EU) AI Act (2024) adopts a risk-based perspective, defining specific types of financial software as “high-risk AI systems” that are subjected to strict transparency, human oversight, accuracy, robustness and cybersecurity provisions (Misheva et al., 2021). US regulators apply existing frameworks but provide additional guidance that indicates increased scrutiny of AI adoption as well as fairness concerns (RFK Human Rights, 2025).

This thematic convergence brings with it an emphasis in many jurisdictions on explainability, fairness and bias mitigation, strong governance, ongoing monitoring, increased documentation, proportionality resort management of third-party AI (Bajracharya et al., 2023; Gandhi et al., 2024).

The disconnection between the regulatory framework and operating guidance remains a barrier. Although regulators express what they want; fair, explainable and robust models, they do not offer much technical specification of how to achieve these goals given the complexity of ML (FCA, 2024).

Table 1 summarizes the major regulatory frameworks addressing ML model risk management, highlighting their key provisions and requirements.

Table 1: Comparison of Major ML Model Risk Management Regulatory Frameworks (2020-2025)

Framework	Jurisdiction	Year	Key Focus Areas	ML-Specific Provisions	Explainability Requirements	Fairness/Bias Requirements
SS 1/23	United Kingdom	2024	First banking-specific AI/ML regulation	Drift monitoring, enhanced documentation, proportionate explainability	High - proportionate to model risk and impact	Bias detection and mitigation required
EU AI Act	European Union	2024	Risk-based approach for high-risk AI systems	Comprehensive AI lifecycle requirements, human oversight, robustness testing	Very High - Article 14 mandates user interpretation	Mandatory fairness assessments for high-risk systems
FCA Literature Review	United Kingdom	Varies; these are ongoing reviews, not single-date acts (e.g., FCA consulted on AI in 2023/2024).	Bias in supervised ML for financial services	Guidance on bias sources, detection methods	Moderate - context-dependent	Extensive focus on algorithmic fairness
GDPR Article 22	European Union	2018 (ongoing)	Rights regarding automated decision-making	Right to contest automated decisions	Implicit requirement for explanation	Limited direct provisions

Source: Compiled from Bank of England (2023), FCA (2024), and regulatory guidance documents

2.2 Validation of Machine Learning Models

Conventional validation frameworks are confronted with significant challenges while validating ML models arising out of algorithmic opacity, high-dimensional feature space, dynamic learning capability and data dependency (Bhatore et al., 2020; Roy & Vasa, 2025).

Conceptual Soundness and Opacity: As ML architectures are made more complex, judging conceptual soundness becomes exponentially harder. The theoretical justification of deep neural networks, not to mention gradient boosting machine or ensemble method, is largely unexplained; their validation heavily relies on empirical superiority rather than fundamental development (Zhang & Yu, 2024). This contrasts with the regulatory demands for theoretical grounded models consistent with established financial theory.

Process Validation and Repeatability: The ML development process is iterative, nonlinear, and often includes steps such as data pre-processing, feature engineering, and hyper parameter tuning. In contrast to typical models with well-defined, specific tasks, ML development can consist of hundreds of iterations of experimentation. Cloud-enabled ML platforms also heighten verifiability challenges by obscuring decision logic and automating model selection using AutoML tools (Robisco & Martínez, 2022).

Barriers to Model Updates and Shift: Since ML models, especially those using online learning approaches, keep adjusting based on new data flow, ongoing monitoring becomes even more important. In this setting, three forms of drift should be monitored: data drift (shifts in input feature distributions), concept drift (changes in the relationships between features and targets), and performance drift (prediction metric degradation). Monthly or quarterly performance evaluations in the traditional sense are insufficient (Tian et al., 2024; Dastile et al, 2020).

Quality and Bias: The ability of ML to work with huge, data sets (of all varieties) mean representativeness is vastly more difficult. More critically, ML's insatiable hunger for data makes it far more likely that any such biases A.I. carries will be deeply embedded. Historical data that captures past biased decisions may include patterns which ML models can pick up and propagate (Mehrabi et al., 2021; Fuster et al., 2022).

Table 2 contrasts the validation challenges between traditional statistical models and ML models, highlighting key differences and emerging solutions.

Table 2: ML Model Validation Challenges vs. Traditional Statistical Models

Validation Aspect	Traditional Statistical Models	Machine Learning Models	Key Challenges for ML	Emerging Solutions
Conceptual Soundness	Clear theoretical foundation; well-understood assumptions	Often empirically driven; limited theoretical justification	Difficulty assessing appropriateness; "black box" nature	Inherently interpretable models (GAMs, EBMs); XAI techniques
Process Verification	Linear development with clear documentation	Iterative with hundreds of experiments; complex pipelines	Reproducibility challenges; AutoML opacity	MLOps practices; version control; experiment tracking
Outcomes Analysis	Standard backtesting; hypothesis testing	Complex performance metrics; multiple validation sets	Model drift; adversarial robustness; fairness assessment	Continuous monitoring; automated testing; fairness audits
Ongoing Monitoring	Periodic recalibration (annual/biennial)	Continuous adaptation; real-time drift	Data drift, concept drift, performance drift	Real-time monitoring systems; automated alerting

Validation Aspect	Traditional Statistical Models	Machine Learning Models	Key Challenges for ML	Emerging Solutions
Data Requirements	Moderate sample sizes; structured data	Large volumes; structured + unstructured data	Representativeness; bias amplification; data quality	Enhanced data governance; bias detection tools; synthetic data
Interpretability	Coefficients directly interpretable	Feature importance often unclear	Regulatory compliance; consumer rights	XAI techniques; model simplification; hybrid approaches
Validation Expertise	Statistics, econometrics, domain knowledge	Statistics + computer science + ML + domain knowledge	Talent scarcity; interdisciplinary requirements	Cross-functional teams; upskilling programs; external experts

Source: Synthesized from Bhatore et al. (2020), Roy & Vasa (2025), Shi et al. (2022), Zhang & Yu (2024)

2.3 Explainable AI: Addressing the Transparency Gap

The biggest block to ML's adoption in a regulated environment bank is explainability. The field of XAI can generally be divided into global explanations (general model performance) and local explanations (individual predictions) (Černevičienė & Kabašinskas, 2024; Ahmed & Iqbal, 2025).

SHAP and LIME: SHAP (Lundberg & Lee, 2017) has become the most popular explainability framework in banking and is based on game theory's Shapley values.SHAP interprets the all-in score as the extent to which each feature contributes to making predictions similar or dissimilar across instances. However, SHAP has critical limitations. Its computational complexity increases exponentially with feature dimensionality. As a result, it is not feasible for very high-dimensional models or real-time applications (Gramegna & Giudici, 2021; Nallakaruppan et al., 2024).

One of the main findings confirms known limitations of SHAP explanations (Cohen et al., 2022; Nallakaruppan et al., 2024), including the fact that they can change dramatically across different background data or trained models, raising questions about their reliability. LIME proposes a solution by locally fitting simpler, interpretable models to approximate complex ones, but is prone to instability, rendering the method unreliable for regulatory purposes (Gramegna & Giudici, 2021; Sowmiya et al., 2024).

Intrinsic Interpretable Models: More recent work has focused on learning models that are interpretable by construction. Generalized Additive Models (GAMs) also approximate the effect of each feature using separate shape functions, which are summed to allow interpretable and visual nonlinearity (Zhang & Yu, 2024; de Lange et al., 2022). Contemporary variants of GAM, including Explainable Boosting Machines (EBMs) and Neural Additive Models (NAMs), embed neural network components, and are interpretable. Empirical evidence from comparative studies shows that these interpretable models can achieve the performance of black box approaches in many designs while providing greater transparency (Kabašinskas & Černevičienė, 2023).

Regulatory Contexts: In regulatory contexts, requirements for model transparency are increasingly explicit. The effect of the EU's GDPR Article 22, which allows people to contest automated decisions, implicitly assumes that automated decisions are explainable. Article 14 of the EU AI Act mandates that high-risk systems should allow users to "correctly interpret the system's output" (Černevičienė & Kabašinskas, 2024). The UK Financial Conduct Authority focuses on proportionate explainability in relation to model risk (FCA, 2024). The issue for banks is deciding what level of explainability is acceptable, given the lack of technical specifications in regulatory guidance.

Table 3 provides a comprehensive comparison of major XAI techniques used in banking applications, evaluating their suitability for regulatory compliance.

Table 3: Explainable AI (XAI) Techniques - Comparative Analysis for Banking Applications

Technique	Type	Scope	Computational Cost	Stability	Key Advantages	Primary Limitations	Banking Suitability
SHAP (Shapley Additive exPlanations)	Post-hoc	Local & Global	High	Moderate	Theoretically grounded; model-agnostic; consistent	Computationally expensive; stability concerns in high dimensions	Moderate - best for offline validation
LIME (Local Interpretable Model-agnostic Explanations)	Post-hoc	Local	Moderate	Low	Fast; model-agnostic; intuitive	High variance; unstable across runs	Low - reliability concerns for regulatory use
Generalized Additive Models (GAMs)	Inherent	Global	Low	High	Interpretable by design; captures nonlinearity	Limited interaction modeling	Very High - regulatory preference
Explainable Boosting Machines (EBMs)	Inherent	Global	Moderate	High	High accuracy with interpretability; pairwise interactions	Relatively new; limited industry adoption	High - emerging best practice
Neural Additive Models (NAMs)	Inherent	Global	Moderate	High	Neural network flexibility with interpretability	Complex implementation	High - promising for complex applications
Partial Dependence Plots (PDPs)	Post-hoc	Global	Moderate	High	Clear visualization	Assumes feature independence	High - good for global

Technique	Type	Scope	Computational Cost	Stability	Key Advantages	Primary Limitations	Banking Suitability
					of feature effects		understanding

Source: Synthesized from Černevičienė & Kabašinskas (2024), Gramegna & Giudici (2021), Nallakaruppan et al. (2024)

2.4 Algorithmic Bias and Fairness

The ability of ML models to reinforce or exacerbate societal biases is a fundamental ethical dilemma and key source of model risk. In contrast with standard models in which discriminatory variables can be directly removed, ML algorithms have the capacity to infer discriminatory patterns from historical data and proxy or complex feature interactions (Mehrabi et al., 2021; Bajracharya et al.,2023)

Sources of Bias: Historical bias is present when the training data replicates earlier discrimination. Bias in representation happens when the training data does not adequately represent some populations. The measurement bias is due to the fact that the measurements of features are more precise in one group than in others (Kheya et al., 2024). And disaggregated analyses continue to demonstrate the presence of racial and ethnic disparities: black and Hispanic applicants for mortgage lending are denied at rates far greater than white applicants with similar credit histories, which translates into some $450 million in extra interest payments per year (Fuster et al., 2022; RFK Human Rights, 2025).

Definitions of Fairness: There is no generally agreed upon definition of fairness, in fact defining fairness in a quantitative sense turns out to be a challenging problem with competing definitions that might be incompatible in the mathematical sense (FCA, 2024; Ford, 2025). On differences between demographic parity, equalized odds, equality of opportunity, predictive parity, individual fairness and counterfactual fairness. It can be proved that, if not in trivial cases, it is mathematically impossible to be simultaneously fair according to multiple criteria.

Bias Mitigation: Bias is mitigated with systematic approaches across the ML pipeline such as preprocessing (weighting samples, deidentification of sensitive attributes), in-processing (fairness-aware regularization, constrained optimization), and post-processing (threshold recalibration). Tools including IBM AI Fairness 360 (Mehrabi et al., 2021), Google What-If Tool (Oguntibeju, 2024) and Microsoft Fairlearn offer a framework for bias detection and mitigation. However, it is not a requirement that more than one fairness metric is included in the study and in fact only one in four studies presents more than one fairness metric, which hinders the comparability between approaches (de Castro Vieira et al., 2025). **Table 4** presents the major fairness metrics used in banking ML applications, including their mathematical definitions, appropriate use cases, and compatibility constraints.

Table 4: Fairness Metrics and Definitions in ML Banking Applications

Fairness Metric	Mathematical Definition	Key Principle	When Most Appropriate	Primary Limitations	Compatibility with Other Metrics
Demographic Parity	$P(\hat{Y}=1\|A=0) = P(\hat{Y}=1\|A=1)$	Similar approval	When equal access is the priority	Ignores base rate differences; may reduce accuracy	Incompatible with most other metrics

Fairness Metric	Mathematical Definition	Key Principle	When Most Appropriate	Primary Limitations	Compatibility with Other Metrics
		rates across groups			
Equalized Odds	P (Ŷ=1|Y=1, A=0) = P (Ŷ=1|Y=1, A=1) AND P (Ŷ=1|Y=0, A=0) = P (Ŷ=1|Y=0, A=1)	Equal TPR and FPR across groups	When accuracy parity is essential	Difficult to achieve; mathematically restrictive	Incompatible with demographic parity
Equality of Opportunity	P (Ŷ=1|Y=1, A=0) = P (Ŷ=1|Y=1, A=1)	Equal TPR across groups	When minimizing false negatives for qualified applicants	Ignores false positive rates	Relaxed version of equalized odds
Predictive Parity	P (Y=1|Ŷ=1, A=0) = P (Y=1|Ŷ=1, A=1)	Equal PPV across groups	When prediction reliability matters across groups	May allow disparate impact	Incompatible with equalized odds when base rates differ
Individual Fairness	Similar individuals receive similar predictions	Consistent treatment of similar cases	When individual equity is paramount	Requires defining similarity metric; computationally complex	Can coexist with some group fairness metrics
Counterfactual Fairness	Predictions unchanged if protected attribute changed	Causal fairness	When causality is critical	Requires causal model; very complex to implement	Theoretical gold standard but impractical

Note: A = protected attribute (e.g., race, gender); Y = true outcome; Ŷ = predicted outcome; TPR = true positive rate; FPR = false positive rate; PPV = positive predictive value

Source: Synthesized from de Castro Vieira et al. (2025), Ford (2025), FCA (2024), Kheya et al. (2024)

2.5 Governance and Organizational Challenges

Successful risk management of ML models is not just a technical challenge but requires organization-wide change, with strong governance, subject matter expertise in specific disciplines and cultural change (Eskandarany, 2024).

Governance needs: MRM governance approaches are not sufficient for changing ML landscapes. More prescriptive guidelines cover well-delineated roles covering model developers, validators and business owners; independent validation outside of development process; oversight by the board and senior management; maintenance of a model inventory log risk appetite frameworks discussing AI/ML specific risks as well as change management policies (Eskandarany, 2024; Roy & Vasa, 2025).

Third-party model governance is particularly challenging, as banks are increasingly reliant on external vendors for machine-learning platforms and pre-trained models, while traditional third-party risk management is constrained by vendors' trade-secret protections over model architectures and training data (Roy & Vasa, 2025).

Talent Development: The talent shortage is one of the greatest hindrances. Conversely, legacy MRM practitioners would have strong foundations in statistics and economics but limited ML experience. On the other end of the spectrum, data scientists will have all sorts of amazing skills around ML and not a clue on banking risks, and regulations requirements. As banks seek to do so, however, they are trying out such approaches as integrated team lay outs, enterprise-level training programs, strategic hiring, academic alliances and MSPs (Eskandarany, 2024).

3. Machine Learning in Banking Applications

Table 5 summarizes the major ML applications in banking, highlighting benefits, risks, validation priorities, and regulatory sensitivity for each domain.

Table 5: ML Applications in Banking - Benefits, Risks, and Validation Priorities

Application Domain	Primary Use Cases	Key Benefits	Primary Risks	Critical Validation Focus	Regulatory Sensitivity
Credit Risk	PD/LGD/EAD modeling; credit scoring; alternative data assessment	1.4-18% AUC improvement; financial inclusion; dynamic risk assessment	Bias amplification; fairness concerns; data quality issues	Fairness audits; stress testing; alternative data validation	Very High - consumer impact
Fraud Detection	Transaction monitoring; AML; identity verification	Real-time detection; pattern recognition; scalability	Imbalanced data; adversarial attacks; false positives	Precision-recall optimization; adversarial robustness; drift monitoring	High - financial crime prevention
Market Risk	Volatility forecasting; option pricing; portfolio optimization	Enhanced predictive accuracy; complex pattern capture	Model drift; black swan events; systemic risk	Stress testing; scenario analysis; continuous retraining	High - systemic implications
Operational Risk	Incident detection; root cause analysis; predictive maintenance	Automation; early warning; efficiency gains	Rare event prediction; data scarcity; long-tailed distributions	Edge case testing; scenario generation; model uncertainty	Moderate - internal focus
Customer Analytics	Churn prediction; next-best-action; personalization	Customer retention; revenue optimization;	Privacy concerns; discrimination	Privacy protection; fairness	Moderate to High -

Application Domain	Primary Use Cases	Key Benefits	Primary Risks	Critical Validation Focus	Regulatory Sensitivity
		experience enhancement	risk; feedback loops	assessment; transparency	consumer protection

Source: Synthesized from Fuster et al. (2022), Heß & Damásio (2025), Dichev et al. (2025), Alamsyah et al. (2025)

3.1 Credit Risk Management

Credit risk is the most mature ML application in banking. ML-based approaches consistently outperform logistic regression with AUC increments of 1.4-18% (Fuster et al., 2022; Heß & Damásio, 2025). Random forests, gradient boosting machines (XGBoost and LightGBM), and neural networks are especially promising for models of probability of default, loss given default, and exposure at default. Examples include using non-traditional data sources for credit scoring in order to assess thin-file customers, thus promising improved financial inclusion while raising fairness concerns (Robisco & Martínez, 2022; Alamsyah et al., 2025).

3.2 Fraud Detection and AML

Pattern recognition is central to ML, and as a result it can be quite efficient in fraud detection and anti-money laundering. Algorithms for anomaly detection detect anomalous activities of transactions. Graph neural networks to analyze transaction networks and uncover money laundering rings (Dichev et al., 2025). The imbalance in fraudulent data poses a validation challenge that needs special treatments, such as SMOTE (Synthetic Minority Over-sampling Technique) and cost-sensitive learning. Real-time deployment needs require low-latency models which results in a trade-off between model complexity and operational performance (Nazareth & Reddy, 2023; Aljunaid et al., 2025).

3.3 Market and Operational Risk

Applications of ML in market risk forecasting include volatility prediction, option pricing and portfolio optimization. The neural networks and the LSTM models memorize long complex temporal dependencies in time series. But the fast paced financial markets present a large amount of model drift that needs to be continuously retrained (Heß & Damásio, 2025). In operational risk, ML can enhance automated detection of incidents and root cause analysis, however long-tailed loss distributions give rise to data scarcity challenges (Dastile et al., 2020).

4. Benefits, Limitations, and Ethical Considerations

4.1 Benefits

ML has its attractions: (1) better predictive accuracy with 1.4-18% AUC improvements leading to meaningful reductions in default rates (Fuster et al., 2022), the ability to process large data and create real-time decisions seamlessly, incorporating alternative data for more well-rounded risk assessment (Alamsyah et al., 2025), being adaptive over time is a key advantage due to an ever-evolving landscape, working at scale reduces human capital requirements so that scarce resources can be directed elsewhere, potential financial inclusion occurs when the technical layer incorporates explicit fairness objectives into design (RFK Human Rights, 2025).

4.2 Limitations and Risks

However, in addition to benefits, ML also raises significant downsides: lack of interpretability which conflicts with regulatory expectations (de Lange et al., 2022), sensitivity to data and quality (Dastile et al., 2020), bias amplification that perpetuates historical discrimination (Mehrabi et al., 2021; Fuster et al., 2022); model drift which necessitates continuous monitoring (Tian et al., 2024), adversarial vulnerability (Aljunaid et al., 2025), complexity and resources intensiveness; overreliance and automation complacency, and finally regulatory uncertainty leading towards compliance challenge (Shi et al., 2022).

4.3 Ethical Considerations

Beyond the technical limitations, ML is also raising serious ethical questions: fairness and discrimination requiring explicit trade-offs between accuracy and fairness (Ford, 2025), privacy and surveillance issues due to fine-grained behavioral data analysis (Eskandarany, 2024), concerns about accountability when ML systems make mistakes in decisions, transparency and consumerist rights for consequential decisions (FCA, 2024), long-term social impact taking feedback loops which can potentially amplify inequality into account (RFK Human Rights, 2025), considerations on human agency in a fully automated decision-making process, environmental sustainability in light of energy footprints.

5. Future Directions and Research Gaps

However, many research gaps still remain:

Standardization of Validation Frameworks: To the best of our knowledge, there is no standardized validation framework available specifically for ML models in banking. Some research needs are in the area of; to include standardized test procedures for different model types, and validation lists to work on ML features (Roy & Vasa, 2025).

Interpretable at scale: XAI methodologies look promising, but they are limited in scalability. There are research gaps in the area of computationally efficient explainability methods for real time applications and schemes to validate explainers (Nallakaruppan et al., 2024; Černevičienė & Kabašinskas, 2024).

Fairness Frameworks: Research must design context-specific fairness frameworks for banking, reconciling regulatory and ethical demands, providing industry consensus on priority fairness metrics for specific use cases (de Castro Vieira et al., 2025; Ford, 2025).

Continuous Monitoring: Automated, ongoing validation mechanisms cannot yet be considered mature. Research requirements also include automated monitoring of systems for real-time drift in a model and methods to alert warnings at the first sign of model erosion (Tian et al., 2024).

Third-Party Risk Management: Banks are increasingly using vendors to provide ML capabilities but the governance mechanisms have yet to catch up. Research requirements can be due diligence frameworks for third party ML providers (Roy & Vasa, 2025).

Harmonization of Regulation: The regulations for managing ML model risk are not harmonized across jurisdictions. There is a need for such research as comparative legal analysis, and policy-oriented research to inform international harmonized standards (Bajracharya 2023; FCA 2024).

6. Conclusion

Embedding machine learning into banking is a seismic shift with tremendous promise and peril. This paper highlights how the pace of rapid ML feature capability development has diverged from traditional model risk management framework adaptation, revealing wide divides between technical capabilities, regulatory requirements and operational practices. Classical MRM formulations continue to be relevant

conceptually, but their extension is non-trivial due to ML's distinctive properties. Although promising interpretability methods have been developed, a tradeoff between performance and explainability continues to exist. Algorithmic fairness is as much a profound ethical imperative as it is a deep technical challenge. Organizational transformation is required for effective ML risk management across governance structures, talent capabilities, and cultural values. The regulatory landscape keeps shifting especially with significant events such as the SS 1/23 from the UK and the EU AI Act. However, significant international fragmentation persists. The industry must now work with regulators and academics to develop standardized validation methods, looking ahead. We need to bolster the advancement of explainability methods. Talent development is a necessary investment for organizations.. Regulators also need to work on global harmonization in a manner that does not prevent flexibility.

When done right, ML adoption can bring greater financial stability, increased financial inclusion, and better operational efficiency. On the other hand, poor risk management exposes financial loss and industry fines, entrenches biased practices, and undermines consumer confidence. As ML continues playing a prominent role in banking systems, robust risk management frameworks must evolve accordingly.

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